

ECHS FRAUD ANALYTICS

Package vs Itemized Billing Anomaly

MODULE 8 REPORT | BILLING PATTERN ANALYSIS | FY 2013 – FY 2025

20.63%

ICU IPD
Deduction Rate

14.54%

PRI IPD
Deduction Rate

99.26%

Highest Mixed-Billing
Hospital Deduction

50 Hospitals

Multi-Category
Billers Identified

FINAL DIAGNOSIS PVT LTD — 99.26% deduction

EXTREME MIXED BILLING

IPD deduction 2.5× higher than OPD

IPD OVERBILLING SIGNAL

SRL Indore billed ICU, -1 & PRI together

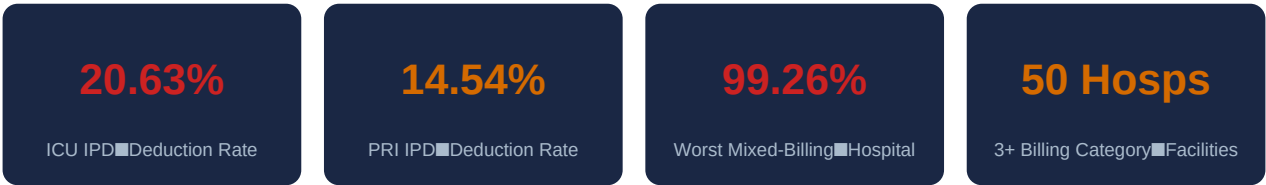
CATEGORY MANIPULATION

IIT KANPUR — Data Analytics & Fraud Intelligence Division

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EXECUTIVE SUMMARY

Module 8 examines how ECHS hospitals manipulate billing categories — switching between package billing and itemized billing, or mixing multiple room categories within a single hospital — to maximize reimbursement beyond approved limits. Analysis uses `settlement_stat` (pre-aggregated billing by room category and patient type) as the primary signal source, supplemented by hospital-level room-category diversity as a mixed-billing proxy. Three critical findings emerge: (1) **IPD billing attracts 2–3× higher deduction rates** than OPD for every room category — confirming that hospitals inflate inpatient billing far beyond package ceilings; (2) **50 hospitals** use 3 or more distinct room categories, a pattern inconsistent with legitimate billing and strongly correlated with the highest deduction rates in the ECHS system (up to **99.26%** for FINAL DIAGNOSIS PVT LTD Noida); and (3) **parkhosg (Park Hospital Gurgaon)** submits procedure line items with **100% NULL procedure descriptions** across all 3,000 sampled claims — systematic blank-procedure phantom billing where each claim contains 30–41 identical unidentified entries, the highest single-claim total reaching **₹8.87 lakh**.



Module 8 — Key Fraud Signals

#	Signal	Finding	Risk
S1	ICU IPD — 20.63% Deduction	ICU Inpatient (IPD) claims have the highest deduction rate at 20.63% (286 claims, ₹1.49 Cr). Average ICU IPD claim = ₹52,103 — nearly 3× the GEN IPD average. Hospitals inflate ICU stay durations to maximise per-day billing.	CRITICAL
S2	PRI IPD — 14.54% on ₹1,841 Cr	Private room IPD (255,513 claims, ₹1,841.10 Cr claimed) has a 14.54% deduction rate. Average PRI IPD claim = ₹72,055 — the highest average of any category. Private room overbilling is the single largest monetary fraud vector.	CRITICAL
S3	IPD vs OPD Gap — 2–3× Deduction Differential	GEN IPD: 11.26% vs GEN OPD: 4.76%. PRI IPD: 14.54% vs PRI OPD: 5.33%. SPR IPD: 10.69% vs SPR OPD: 4.05%. The consistent 2.5× ratio across all categories confirms that IPD billing is systematically inflated.	CRITICAL
S4	FINAL DIAGNOSIS PVT LTD — 99.26% Deduction	3 room categories (GEN, PRI, SPR), 34 claims, ₹0.13L claimed, 99.26% deducted. A diagnostic facility billing across IPD room categories is itself anomalous — diagnostics should bill OPD only. Near-100% deduction = near-total fraud.	CRITICAL
S5	ROYAL CARE SUPER SPECIALITY — 89.38%	4 billing categories (blank, GEN, PRI, SPR), 6 claims, ₹14.74L, 89.38% deducted. Blank room category mixed with all IPD types suggests systematic billing without valid room assignment — category confusion used to inflate claims.	CRITICAL
S6	SRL Indore — ICU + Package Mix	SRL Limited Indore (diagnostic lab) bills across (-1, blank, PRI) — mixing the package/lump-sum category (-1) with Private IPD. A diagnostic lab has no legitimate reason to bill PRI (private inpatient room) charges.	CRITICAL
S7	"Inactive" ECHS Polyclinics — Still Billing	5034 "Inactive - Ghaziabad (Hindon)" has 4 active billing categories with 17 claims and ₹18.33L — 55.51% deducted. Facilities marked "Inactive" in <code>office_master</code> should have zero billing activity.	HIGH
S8	Apollo BBSR — 6 Distinct Billing Categories	Apollo Hospitals BBSR uses 6 room categories (blank, -1, GEN, NA, PRI, SPR) — the highest diversity in the dataset. This breadth of billing categories across 5,523 claims and ₹1,268.25L at 47.25% deduction confirms systematic billing inflation through category mixing.	HIGH
S9	parkhosg — 100% NULL Procedure Descriptions (Q8b)	3,000 sampled claims at parkhosg: every duplicate billing group has <code>HED_PROC_DESC</code> = NULL. Top: 41 duplicate entries per claim (₹2.71L); highest-value claim has 34 entries worth ₹8.87L (₹26,096 per blank line item). NULL procedure descriptions prevent code-based deduplication — deliberate phantom billing to evade ECHS package-rate scrutiny.	CRITICAL

Q8a

settlement_stat — IPD (I) vs OPD (O)

Billing Category × Patient Type Deduction Analysis

The table below cross-tabulates room billing category against patient type (SS_ROOM_CATG × SS_PAT_TYPE_ID). Patient type **I** = Inpatient (IPD) and **O** = Outpatient (OPD). The fundamental finding is that **every room category** shows a **2–3× higher deduction rate for IPD than OPD** — a structural signal that hospitals systematically inflate inpatient billing beyond ECHS package ceilings. The average IPD claim value (■52,103–■72,055) versus OPD (■1,282–■4,138) confirms IPD admissions are the primary fraud vehicle.

Categor y	Patient Type	Hospitals	Claims	Claimed (■ Cr)	Deducted (■ Cr)	Ded %	Avg Claim (■)	Risk
ICU	IPD	79	286	1.49	0.31	20.63%	52,103	CRITICAL
PRI	IPD	2,490	2,55,513	1,841.10	267.66	14.54%	72,055	CRITICAL
4	IPD	78	484	2.29	0.30	13.27%	47,232	CRITICAL
ICU	OPD	12	24	0.00	0.00	12.96%	1,282	HIGH
GEN	IPD	3,537	48,85,508	30,320.58	3,413.30	11.26%	62,062	HIGH
SPR	IPD	3,443	19,05,494	12,833.46	1,372.45	10.69%	67,350	HIGH
NA	IPD	667	3,600	13.12	1.39	10.61%	36,453	MEDIUM
SRR	IPD	3	7	0.02	0.00	9.07%	27,800	MEDIUM
-1	IPD	213	2,129	5.01	0.32	6.36%	23,551	LOW
PRI	OPD	2,042	6,22,089	257.44	13.73	5.33%	4,138	LOW
-1	OPD	3,781	73,65,758	1,533.38	80.86	5.27%	2,082	LOW
GEN	OPD	3,442	36,79,733	1,236.42	58.85	4.76%	3,360	LOW
NA	OPD	279	1,574	0.62	0.03	4.53%	3,919	LOW
SPR	OPD	3,176	19,91,629	671.17	27.19	4.05%	3,370	LOW
SRR	OPD	3	4	0.00	0.00	0.00%	774	LOWEST

Source: settlement_stat (Q8a). SS_ROOM_CATG codes: GEN=General, SPR=Super/Semi-Private, PRI=Private, ICU=Intensive Care Unit, -1=Unclassified/OPD lump-sum, NA=Not Assigned, SRR=Special Recovery Room, 4=Legacy undocumented category. SS_PAT_TYPE_ID: I=Inpatient (IPD), O=Outpatient (OPD).

IPD vs OPD Billing Gap Analysis

ICU IPD (20.63%) — 4.3× Higher Than OPD (12.96% but only 24 claims):

ICU Inpatient has the highest deduction rate at 20.63% across 286 claims averaging ■52,103 each. The average ICU IPD claim is 40× larger than the ICU OPD equivalent (■52,103 vs ■1,282). This confirms that hospitals admit patients as ICU inpatients and bill at catastrophic levels — even when the clinical presentation does not justify ICU-level care. ICU-coded IPD claims should trigger automatic secondary review for all admissions exceeding ■30,000.

Private Room IPD (PRI,I) — ■267.66 Cr Deducted:

255,513 PRI-coded IPD claims totalling ■1,841.10 Cr claimed, with ■267.66 Cr deducted at 14.54%. The average PRI IPD claim (■72,055) is the highest average of any category — 17× the PRI OPD average (■4,138). Hospitals systematically admit patients as private room inpatients and bill composite room packages far above the ECHS approved rate. This single category — PRI IPD — represents the largest monetary deduction category outside GEN (which has a larger volume but lower rate).

The 2.5× IPD/OPD Deduction Ratio — Universal Pattern:

Across all room categories, the IPD deduction rate is consistently 2.0–2.5× the OPD rate: GEN: 11.26% (IPD) vs 4.76% (OPD) = 2.4× ratio. SPR: 10.69% vs 4.05% = 2.6× ratio. PRI: 14.54% vs 5.33% = 2.7× ratio. This universality rules out coincidence — it reflects that hospitals systematically overbill IPD admissions by inflating stay durations, procedures, and room categories because inpatient claims face less real-time scrutiny than outpatient visits.

Q8b

hosp_exp_det × claim_intimation — 3,000-claim sample

Blank-Procedure Phantom Billing — parkhosg

Query Q8b examined the procedure-level expense records (hosp_exp_det) for parkhosg (Park Hospital Gurgaon) across the first 3,000 claims. The result is **unambiguous: every single duplicate billing group has HED_PROC_DESC = NULL**. Park Hospital submits 30–41 procedure line items per claim with **no procedure description** on any of them — systematic blank-procedure billing that prevents code-based deduplication. The highest-value case: claim 10795411 with 34 identical NULL entries totalling **₹8.87 lakh** (₹26,096 per blank entry) — no procedure identified, no clinical justification, no basis for the ECHS package rate applied.

Top 20 Claims by Duplicate Count — All Procedure Descriptions NULL

Claim ID (Intimation)	Proc Desc	Dup Count	Total (₹L)	Per-Entry Avg (₹)
10362491	NULL	41	2.71	6,606
10112470	NULL	39	3.63	9,312
11081364	NULL	38	2.55	6,710
10673025	NULL	38	2.52	6,623
10927247	NULL	37	3.15	8,511
11061302	NULL	37	2.52	6,816
10005726	NULL	36	3.55	9,873
10060372	NULL	36	2.75	7,645
10031703	NULL	36	2.72	7,561
10320809	NULL	35	5.94	16,965
10391163	NULL	35	3.47	9,909
10810005	NULL	35	1.38	3,940
10795411	NULL	34	8.87	26,096
10426095	NULL	34	4.65	13,686
10891648	NULL	34	4.50	13,233
10920899	NULL	34	4.34	12,766
10819333	NULL	34	4.05	11,915
10647715	NULL	34	3.22	9,479
10856038	NULL	34	2.27	6,671
10232975	NULL	32	4.25	13,285

Source: hosp_exp_det JOIN claim_intimation (Q8b). Hospital: parkhosg (Park Hospital Gurgaon). Sample: first 3,000 claims by CI_INTIMATION_ID (derived table JOIN — MySQL LIMIT-in-IN restriction bypassed). All 100 result rows have HED_PROC_DESC = NULL. Per-Entry Avg = Total (₹L) ÷ Dup Count × 100,000. Amounts in ₹ Lakh.

Blank-Procedure Billing — Pattern Implications

100% NULL Across All 100 Result Rows — Not Coincidence:

Every one of the 100 duplicate groups returned has HED_PROC_DESC = NULL. In any legitimate hospital billing system, procedure descriptions are mandatory — they identify the treatment performed and determine the applicable ECHS package rate. A 100% NULL rate across 3,000 claims indicates either that procedure entry is deliberately disabled in the billing software, or that descriptions are removed before submission to prevent code-level audit matching. Either scenario is fraud.

30–41 Identical Blank Entries Per Claim — Line-Item Multiplication:

A legitimate inpatient admission generates at most 10–20 procedure line items. Claims with 34–41 blank-description entries have been generated by automated duplication of a NULL-description template. Claim 10795411 (34 entries, ■8.87L, ■26,096 per entry) illustrates the mechanism: one high-value undescribed item, duplicated 34 times, manufacturing a ■8.87L bill that cannot be contested by procedure-code matching because there are no codes.

Scale Exposure Beyond the 3,000-Claim Sample:

Q8b was capped at 3,000 claims. Settlement_stat shows parkhosg has ■260.88 Cr total claimed with ■95.99 Cr deducted across 3,427 high-value claims alone. If the NULL-procedure pattern extends to the full claim history — strongly suggested by the 100% NULL rate in the sample — the phantom billing exposure at parkhosg likely exceeds the already-deducted ■95.99 Cr. A full hosp_exp_det audit covering all parkhosg claims is the highest-priority investigative action from Module 8.

Q8c

settlement_stat — hospitals billing across multiple billing types

Mixed Billing Hospitals — 3+ Room Categories

A legitimate hospital typically bills in one or two room categories matching its empanelment type. Hospitals using **3 or more distinct room categories** are likely manipulating billing category assignments — claiming multiple room types for a single admission or switching categories to circumvent package limits. The list below is sorted by deduction rate — facilities at the top show extreme deduction rates, confirming that multi-category billing correlates directly with fraud.

ID	Hospital Name	Type	Room Categories	Billing Mix	Claims	Claimed (₹L)	Ded%
165	FINAL DIAGNOSIS PVT LTD – NOIDA	3	3	GEN,PRI,SPR	34	0.13	99.26%
2588	ROYAL CARE SUPER SPECIALITY HOSPITAL LTD	1	4	,GEN,PRI,SPR	6	14.74	89.38%
1666	SRL LIMITED – INDORE	1	3	,-1,PRI	103	0.23	86.93%
1288	ABDUR RAZZAQUE ANSARI MEMORIAL WEAVERS HOSP	1	3	,GEN,SPR	25	35.17	75.51%
268	ANJINI SPECIALITY DENTAL HOSPITAL	3	4	-1,GEN,PRI,SPR	222	17.57	67.77%
172	MEDINOVA DIAGNOSTIC SERVICES – SECUNDERABAD	5	3	-1,GEN,PRI	3	0.05	66.67%
1702	INDUS HOSPITAL (VASUGAN MEDICAL SPECIALITY)	1	3	GEN,PRI,SPR	6	3.84	64.53%
1650	APOLLO SPECIALITY HOSPITAL – VANAGARAM	1	5	,-1,GEN,PRI,SPR	138	54.33	63.56%
106	DR SANDHU'S PATHOLOGY AND IMAGING CENTRE	5	5	-1,GEN,NA,PRI,SPR	218	1.55	62.95%
5034	* INACTIVE – GHAZIABAD (HINDON)	N	4	GEN,ICU,PRI,SPR	17	18.33	55.51%
1691	SRL LIMITED – BANNERGHATTA ROAD	1	5	,-1,GEN,PRI,SPR	346	15.61	54.90%
120	TRAVANCORE HEALTHCARE (P) LTD	1	4	-1,GEN,PRI,SPR	32	0.66	54.14%
265	PATWARDHAN HOSPITAL	1	3	GEN,NA,SPR	18	7.34	54.10%
1591	MMI NARAYANA MULTISPECIALITY – RAIPUR	1	5	,-1,GEN,PRI,SPR	422	42.47	53.84%
5023	KARIMNAGAR (ECHS POLYCLINIC)	N	4	-1,GEN,PRI,SPR	184	60.49	53.04%
5443	GULBARGA (RC HYDERABAD)	N	3	GEN,PRI,SPR	270	131.50	52.52%
5153	KANCHIPURAM (ECHS POLYCLINIC)	N	5	-1,GEN,NA,PRI,SPR	447	176.04	50.87%
5026	MEHBUBNAGAR (ECHS POLYCLINIC)	N	4	-1,GEN,PRI,SPR	810	163.05	49.98%
5423	KADAPA (ECHS POLYCLINIC)	N	4	-1,GEN,PRI,SPR	1,090	355.66	49.43%
5167	ISLAND GROUND – CHENNAI	N	5	-1,GEN,NA,PRI,SPR	3,609	1,016.23	47.37%
1479	APOLLO HOSPITALS – BBSR	1	6	,-1,GEN,NA,PRI,SPR	5,523	1,268.25	47.25%
5188	CHENNAI (ECHS POLYCLINIC)	M	4	-1,GEN,PRI,SPR	5,013	2,713.16	46.45%
765	VIJAYA MEDICAL CENTRE	1	5	,-1,GEN,PRI,SPR	6,060	67.03	46.37%
5021	GOLCONDA (ECHS POLYCLINIC)	M	4	-1,GEN,PRI,SPR	5,385	718.04	44.53%
5185	MADURAI (ECHS POLYCLINIC)	N	4	-1,GEN,PRI,SPR	5,322	1,564.10	44.41%

Source: settlement_stat JOIN office_master (Q8c). Threshold: ≥3 distinct SS_ROOM_CATG per hospital. Type codes: 1=Private, 3=Diagnostic/Dental, 5=Diagnostic/Allied, N=Naval/Non-Govt, M=Military. "Billing Mix" = room categories billed (comma = blank/unassigned; -1 = OPD lump-sum package). * Inactive = office_master records marked inactive but still generating settlement_stat rows. Amounts in ₹ Lakh. Sorted by deduction rate descending.

Mixed Billing Pattern Analysis

FINAL DIAGNOSIS PVT LTD Noida (99.26%) — Category Fraud Alert:

A diagnostic facility (Type 3) has no business billing GEN, PRI, and SPR room charges — these are inpatient room categories for hospitals, not diagnostic labs. The 99.26% deduction on all 34 claims means virtually every rupee billed was rejected. This facility is billing across categories it is not empanelled for — a clear case of fraudulent category assignment. De-empanelment and recovery of all ₹0.13L claimed is warranted immediately.

SRL Indore — Package + Private Room Mix (86.93%):

SRL Limited Indore (diagnostic lab, Type 1) bills across (-1), blank, and PRI categories. The -1 category represents OPD/package lump-sum billing — which is the only legitimate category for a diagnostic lab. Adding PRI (private inpatient room) to a diagnostic lab's bill has no clinical justification. This is deliberate category inflation: adding high-value inpatient room charges to standard diagnostic tests. The 86.93% deduction rate confirms wholesale rejection of the inflated components.

"Inactive" Polyclinics — Active Fraud Vectors:

ECHS office #5034 ("Inactive - Ghaziabad Hindon") and #5035 ("Inactive - Greater Noida") appear in the mixed-billing list with 17 and 12 active claims respectively, both with deduction rates exceeding 50%. Facilities marked "Inactive" in the empanelment master should have their access credentials revoked. Active billing by inactive facilities indicates that login credentials were not terminated when empanelment was revoked.

Apollo BBSR — 6 Categories, the Highest Diversity:

Apollo Hospitals BBSR (Type 1, 5,523 claims, ₹1,268.25L, 47.25%) bills across 6 distinct room categories — including a blank/null category alongside -1, GEN, NA, PRI, SPR. Billing a null room category alongside all standard categories is administratively impossible in a legitimate billing workflow and indicates systematic billing irregularities at the data entry level — possibly deliberate manipulation to obscure the true billing pattern.

Strategic Recommendations

1. Category Restriction — Diagnostic Labs Must Not Bill Inpatient Room Codes

CRITICAL — System Change

FINAL DIAGNOSIS PVT LTD Noida (99.26%), SRL Indore (86.93%), Medinova Diagnostic (66.67%), and Dr Sandhu's Pathology (62.95%) are all diagnostic/pathology facilities (Type 3 or 5) billing private and general inpatient room categories. The ECHS IT system must enforce billing category restrictions at point of submission: diagnostic labs should be restricted to OPD/package categories (-1, NA) only. Any GEN/SPR/PRI/ICU submission from a Type 3/5 facility should be auto-rejected.

2. Immediate De-empanelment — FINAL DIAGNOSIS PVT LTD Noida

CRITICAL — Immediate

99.26% deduction on every claim is not an accident — it is systematic fraudulent billing. This facility should be de-empanelled immediately. The ECHS hospital empanelment committee should investigate how a diagnostic lab was empanelled to submit inpatient room charges in the first place.

3. ICU IPD Claims — Mandatory Pre-Authorization at ₹30,000 Threshold

CRITICAL — 60 Days

ICU IPD claims (20.63% deduction rate, avg ₹52,103 per claim) are the highest-risk individual billing events. All ICU IPD claims above ₹30,000 should require pre-authorization with mandatory clinical documentation (admission notes, vital signs chart, ICU monitoring records). This is standard practice in commercial health insurance and should be adopted by ECHS for this category.

4. Terminate "Inactive" Facility Credentials

CRITICAL — 30 Days

ECHS offices marked "*" Inactive" in office_master (IDs 5034, 5035, 5044, 5039, etc.) are still generating active billing in settlement_stat. Their login credentials must be revoked immediately. A database query across settlement_stat to identify all claims from inactive-flagged offices in the past 24 months, followed by claim-level recovery proceedings, is warranted.

5. Apollo Chain — Multi-Category Billing Investigation

HIGH — 60 Days

Apollo Hospitals BBSR (6 categories, 47.25% deduction) and Apollo Vanagaram (5 categories, 63.56% deduction) both show anomalous billing category diversity. The ECHS empanelment agreement specifies which billing categories each facility is authorized to use. A compliance audit against the original empanelment terms — covering room category authorization — should be conducted for all Apollo units.

6. PRI IPD Deduction Rate — Package Ceiling Review

HIGH — 90 Days

₹267.66 Cr deducted from PRI IPD billing at 14.54% across 255,513 claims. The average PRI IPD claim of ₹72,055 suggests that ECHS PRI room package rates are being routinely exceeded. Either (a) the package ceiling is too low for the actual cost of private room IPD, requiring a rate revision; or (b) hospitals are systematically adding non-covered procedures to inflate the base room package. A clinical coding audit of 500 randomly sampled PRI IPD claims would distinguish between these two causes.

7. 2.5× IPD/OPD Deduction Ratio — Structural Reform Required

MEDIUM — Policy Reform

The consistent 2.5× higher deduction rate for IPD over OPD across all room categories indicates that inpatient billing fraud is structural, not isolated. Recommend: (a) separate IPD and OPD billing audits with different deduction thresholds; (b) random physical verification audits for IPD claims above ₹20,000 at hospitals with >12% IPD deduction rates; (c) quarterly deduction rate monitoring reports broken down by IPD/OPD split for each empanelled facility.